

Somnath Pandit

Software Engineer | Backend & IoT Systems

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SUMMARY

Software Engineer experienced in building **backend services**, **full-stack applications**, and **IoT systems** that connect software with real-world devices. Skilled in Node.js, Java/Spring Boot, C/C++, React, embedded Linux, MQTT, and WebSockets. Hands-on experience with Raspberry Pi/CM4 systems, hardware telemetry, network management, and production deployments. Comfortable designing scalable APIs, microservices, real-time data pipelines, and CI/CD workflows.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Nov 2022 – Jun 2026

[University of Kalyani](#)

Kalyani, India

- **Leadership:** Academic General Secretary (AGS), Department of CSE (2024)

SKILLS

Languages: Java 17, TypeScript/JavaScript, C/C++, Python, SQL

Backend & Frameworks: Node.js, Express.js, Spring Boot, FastAPI, C++ (Crow, Boost.Asio), REST APIs, MQTT, WebSockets

Frontend: React 19, Next.js, ReactFlow, Vite, Tailwind CSS, Material UI, Android

Databases & Storage: MySQL, PostgreSQL (Native pg), SQLite

Systems & Infrastructure: Linux (Ubuntu 24.04 LTS), Nginx, PM2, Docker, ESP32 Dev Kit V1, GSM EC200, Git

EXPERIENCE

Software Engineer

12/2025 – 06/2026

[Lilac Inc](#)

Kolkata, India

- **Audlog Pro Voicelogger:** Built a production audio logging server (Node.js, React, MySQL) on Ubuntu. Configured Nginx, PM2, and custom filesystem pipelines for large-scale `.opus` storage.
- **IoT Telemetry & LC Gate:** Engineered a real-time railway monitoring system using the ESP32 Dev Kit V1, MQTT, and WebSockets. Integrated EC200 GSM modules for remote web-based network management.
- **Hardware Metrics (C++):** Developed Linux microservices via C++ (Crow, Boost.Asio) to expose CPU, disk, and network states through REST APIs.

PROJECTS

[Visual DAG Execution Platform](#) | *React, ReactFlow, FastAPI, Python*

[Live](#) | [GitHub](#)

- Built a visual node-based editor for designing and validating complex AI data pipelines directly in the browser.
- Implemented Kahn's Topological Sort Algorithm on the FastAPI backend to detect graph cycles and validate Directed Acyclic Graphs (DAGs) in $O(V + E)$ time.

[NexHire ATS Ecosystem](#) | *React 19, Node.js, PostgreSQL, Zod*

[Live](#) | [GitHub](#)

- Engineered a production-ready Applicant Tracking System featuring a decoupled architecture separating client-side rendering from backend data APIs.
- Utilized direct PostgreSQL queries via the native `pg` client for zero-ORM performance overhead, enforcing strict schema validation with Zod and JWT role-based access control.

[GA - Scheduler \(NP-Hard Solver\)](#) | *Next.js, Zustand, Vite PWA*

[Live](#) | [GitHub](#)

- Authored an offline-first Progressive Web App to solve the University Course Timetabling Problem entirely client-side.
- Designed a matrix-centric Genetic Algorithm avoiding 1D chromosome destruction, achieving 42% faster convergence through targeted heuristic mutation and vectorized $O(1)$ constraint lookups.

[dev.restaurant Ecosystem](#) | *Spring Boot, React, Android, Microservices*

[Live](#) | [GitHub](#)

- Architected a highly scalable 4-tier microservice system for seamless food delivery operations.
- Integrated web dashboards and mobile touchpoints to synchronize real-time order processing and restaurant management.

[ZeroProof: Anonymous Feedback System](#) | *Node.js, TypeScript, React*

[Live](#) | [GitHub](#)

- Architected an anonymous feedback system using RSA Blind Signatures to cryptographically decouple voter identities from submission receipts.
- Engineered OTP invite workflows, token issuance, and admin verification dashboards via secure REST APIs.